

Woong-Hee Shin is an Associate Professor in the Department of Biomedical Informatics at Korea University College of Medicine. He received his B.S. and Ph.D. in Chemistry from Seoul National University, where his doctoral research focused on the development of computational methods for protein–ligand docking. He subsequently conducted postdoctoral research at Purdue University in the Kihara Bioinformatics Laboratory before joining academia.

His research focuses on understanding and predicting biomolecular interactions by integrating physical chemistry, structural bioinformatics, and artificial intelligence. In particular, he develops computational approaches for protein–ligand docking, structure-based virtual screening, binding affinity prediction, and AI-driven drug discovery. His recent work includes AlphaFold2-based multi-state modeling, graph neural network approaches for protein–ligand interaction prediction, and deep learning methods for million-scale virtual screening. His research aims to improve the accuracy, efficiency, and scalability of computational drug discovery.